Newman-Penrose Formalism Calculations

Stephani et al., Equation (28.21)

Metric

$$ds^2 = -\frac{2\,(Kr - 2\,m)}{r}\mathrm{d}u \otimes \mathrm{d}u - \mathrm{d}u \otimes \mathrm{d}r - \mathrm{d}r \otimes \mathrm{d}u + \left(\frac{16\,r^2}{K^2x^4 + K^2y^4 + 8\,Kx^2 + 2\,(K^2x^2 + 4\,K)y^2 + 16}\right)\mathrm{d}x \otimes \mathrm{d}x + \left(\frac{16\,r^2}{K^2x^4 + K^2y^4 + 8\,Kx^2 + 2\,(K^2x^2 + 4\,K)y^2 + 16}\right)\mathrm{d}y \otimes \mathrm{d}y$$

Coordinates

$$(x^\mu) = (u, r, x, y)$$

Null Tetrad

Covariant components

$$l = (1)\ du$$

$$n = \left(\frac{Kr - 2\,m}{r}\right)\ du + (1)\ dr$$

$$m = \left(\frac{2\,\sqrt{2}r}{Kx^2 + Ky^2 + 4}\right)\ dx + \left(\frac{2i\,\sqrt{2}r}{Kx^2 + Ky^2 + 4}\right)\ dy$$

$$\overline{m} = \left(\frac{2\,\sqrt{2}r}{Kx^2 + Ky^2 + 4}\right)\ dx + \left(-\frac{2i\,\sqrt{2}r}{Kx^2 + Ky^2 + 4}\right)\ dy$$

Contravariant components

$$l = (-1)\ dr$$

$$n = (-1)\ du + \left(\frac{Kr - 2\,m}{r}\right)\ dr$$

$$m = \left(\frac{\sqrt{2}\left(Kx^2 + Ky^2 + 4\right)}{8\,r}\right)\ dx + \left(\frac{\sqrt{2}\left(i\,Kx^2 + i\,Ky^2 + 4i\right)}{8\,r}\right)\ dy$$

$$\overline{m} = \left(\frac{\sqrt{2}\left(Kx^2 + Ky^2 + 4\right)}{8\,r}\right)\ dx + \left(\frac{\sqrt{2}\left(-i\,Kx^2 - i\,Ky^2 - 4i\right)}{8\,r}\right)\ dy$$

Directional Derivatives

$$D = -\partial_r$$

$$\Delta = -\partial_u + \frac{Kr - 2\,m}{r}\,\partial_r$$

$$\delta = \frac{\sqrt{2}\left(Kx^2 + Ky^2 + 4\right)}{8\,r}\,\partial_x + \frac{\sqrt{2}\left(i\,Kx^2 + i\,Ky^2 + 4i\right)}{8\,r}\,\partial_y$$

$$\overline{\delta} = \frac{\sqrt{2}\left(Kx^2 + Ky^2 + 4\right)}{8\,r}\,\partial_x + \frac{\sqrt{2}\left(-i\,Kx^2 - i\,Ky^2 - 4i\right)}{8\,r}\,\partial_y$$

Spin Coefficients

$$\rho = \frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{K r - 2\,m}{r^2}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = \frac{\sqrt{2} K (x - i\,y)}{8\,r}$$

$$\beta = -\frac{\sqrt{2} K (x + i\,y)}{8\,r}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = -\frac{m}{r^2}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{K}{4\,r^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = -\frac{K}{12\,r^2}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{K r - 12\,m}{6\,r^3}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"